



Miss Mehwish: 03171491481

Sta632

Final Term (Live Quiz)

PAID VU LMS HANDLING by Mam Mehwish

03184148783

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STA632 - Sampling Techniques (Quiz-3) Quiz Start Time: 09:52 PM, 05 January 2025

Question # 4 of 10 (Start time: 09:53:01 PM, 05 January 2025) Total Marks: 1

The below expression for variance of mean estimator under ranked set sampling is:

Select the correct option

☒ $V(\bar{X}^*) = \frac{1}{nk} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$

☐ $V(\bar{X}^*) = \frac{1}{k} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$

☐ $V(\bar{X}^*) = \frac{1}{nk} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{m} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$

☐ $V(\bar{X}^*) = \frac{1}{k^2} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$

[Click to Save Answer & Move to Next Question](#)

Subject Code: STA632 - Sampling Techniques (Quiz-4)

When the population size is what size, it is recommended to use the entire population instead of a sample?

100 or greater

1000 or greater

100 or less ✓



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1000 or less

What is a potential advantage of using the regression estimator in survey sampling?

It provides unbiased estimates even in the presence of non-response.

It is less sensitive to outliers in the data.

It is computationally simpler than other estimators.

It often results in more efficient estimates when the auxiliary variable is correlated with the variable of interest. ✓

In R, which function is commonly used to perform probability proportional to size sampling?

ppsampling()

sample() ✓

psize()

stratified()

In survey sampling, what is the use of ratio estimator?

Estimating population mean ✓

Estimating population proportions

Which of the following is a common method used for estimating the population mean using an auxiliary variable in survey sampling?

Culster random sampling

Simple random sampling

Stratified random sampling

Ratio estimation ✓

In sampling methods, what does "probability proportional to size" mean?



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The probability of selecting a unit is directly proportional to its size or measure ✓

Sampling units are chosen randomly from the population

Sampling units are selected based on their alphabetical order

All units in the population have an equal chance of being selected

When Grant estimated the population of England:

1962

1667

1952

1662 ✓

If the ratio estimator is unbiased, what can we conclude about its expected value?

It is equal to the sample mean.

It is equal to the true population parameter. ✓

It is always equal to zero.

It is equal to the true population parameter.

What condition must be satisfied for the ratio estimator to be unbiased?

The ratio of the population means must be equal to the ratio of the sample means. ✓

The auxiliary variable and the variable of interest must be independent.

The product of the population means must be equal to the product of the sample means.

The auxiliary variable must be perfectly correlated with the variable of interest.

Auxiliary variable is also called:



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independentVariable

sampling Variable

supporting variable ✓

dependentVariable

What condition must be satisfied for the ratio estimator to be unbiased?

The ratio of the population means must be equal to the ratio of the sample means. ✓

The auxiliary variable must be perfectly correlated with the variable of interest.

The auxiliary variable and the variable of interest must be independent.

The product of the population means must be equal to the product of the sample means.

When the population size is what size, it is recommended to use the entire population instead of a sample?

100 or greater

100 or less ✓

1000 or greater

1000 or less

What is a potential advantage of using the ratio estimator in survey sampling?

It is computationally simpler than other estimators.

It is less sensitive to outliers in the data.

It provides unbiased estimates even in the presence of non-response.

It often results in more efficient estimates when the auxiliary variable is correlated with the variable of interest. ✓

In survey sampling, what is the use of ratio estimator?



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Estimating population mean ☒

Estimating population proportions

If the ratio estimator is unbiased, what can we conclude about its expected value?

It is equal to the sample mean.

It is equal to the true population parameter. ☒

It is always equal to zero.

When Grant estimated the population of England:

1952

1667

1662 ☒

1962

In R, which function is commonly used to perform probability proportional to size sampling?

sample() ☒

psize()

stratified()

ppsampling()

Which of the following is a common method used for estimating the population mean using an auxiliary variable in survey sampling?

Stratified random sampling

Ratio estimation ☒

Cluster random sampling



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Simple random sampling

Auxiliary variable is also called:

independentVariable

dependentVariable

supporting variable ☒

sampling Variable

What is a potential advantage of using the regression estimator in survey sampling?

It is less sensitive to outliers in the data.

It provides unbiased estimates even in the presence of non-response.

It often results in more efficient estimates when the auxiliary variable is correlated with the variable of interest. ☒

It is computationally simpler than other estimators.

Question #1:

Farmer Joe separates his farm into 10 regions. He then randomly selected 5 trees from each region to estimate the number of apples produced on his apple tree farm. This is _____ sampling.

- Cluster
- Systematic
- Simple
- ☒ Stratified

Question #2:

Mrs. Trahan samples her class by selecting 5 girls and 7 boys. This type of sampling is called?

- Simple
- Systematic



- If the cluster sizes are unequal, the mean of cluster size is equal to:



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Question #6:

The population Mean in Case of Unequal Cluster Sizes is:

- False
- [✓] True

Question #7:

In R, which function is commonly used to perform probability proportional to size sampling?

- psize()
- sample()
- stratified()
- [✓] ppsampling()

Question #8:

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

- [✓] Random sampling
- Non-Random sampling

Question #9:

Farmer Joe randomly picks 100 trees using a random number generator to estimate the number of apples produced by his apple trees. This is _____ sampling.

- [✓] Simple
 - Cluster
 - Stratified
 - Systematic
-



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Question #10:

In sampling methods, what does "probability proportional to size" mean?

- Sampling units are chosen randomly from the population
- [✓] The probability of selecting a unit is directly proportional to its size or measure
- All units in the population have an equal chance of being selected
- Sampling units are selected based on their alphabetical order

Question 1:

The population Mean in Case of Unequal Cluster Sizes is:

$$\sum_{i=1}^N M_i y_i - iM'$$

- True ✓
- False

Question 2:

A sample which is free from bias is called:

- Biased
- Unbiased ✓
- Positively biased
- Negatively biased

Question 3:

In sampling methods, what does "probability proportional to size" mean?

- All units in the population have an equal chance of being selected
- Sampling units are chosen randomly from the population
- Sampling units are selected based on their alphabetical order



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- The probability of selecting a unit is directly proportional to its size or measure ☒

Question 4:

Horvitz and Thompson introduced an unbiased estimator in:

- 1697
- 1967 ☒
- 1662
- 1952

Question 5:

Farmer Joe separates his farm into 10 regions. He then randomly selected 5 trees from each region to estimate the number of apples produced on his apple tree farm. This is _____ sampling.

- Cluster
- Systematic
- Stratified ☒
- Simple

Question 6:

Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?

- Non-Random sampling
- Random sampling ☒

Question 7:

The below expression for variance of mean estimator under ranked set sampling is:

$$V(\bar{X}^*) = \frac{1}{n} \sum_{i=1}^k E(X(i)^* - \bar{X}^*)^2 - \sum_{i=1}^k (X(i)^* - \bar{X}^*)^2$$

- $\frac{1}{k} \sum_{i=1}^k E(X(i)^* - \bar{X}^*)^2 - \frac{1}{k} \sum_{i=1}^k (X(i)^* - \bar{X}^*)^2$



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- $\frac{1}{n} \sum_{i=1}^n E(X(i) - \bar{X})^2 = \frac{1}{m} \sum_{i=1}^m k(X(i) - \bar{X})^2$
- $\frac{1}{n} \sum_{i=1}^n E(X(i) - \bar{X})^2 = \sum_{i=1}^n k(X(i) - \bar{X})^2$ ✓
- $\frac{1}{k} \sum_{i=1}^k E(X(i) - \bar{X})^2 = \sum_{i=1}^k k(X(i) - \bar{X})^2$

Question 8:

Farmer Joe randomly picks 100 trees using a random number generator to estimate the number of apples produced by his apple trees. This is _____ sampling.

- Systematic
- Stratified
- Cluster
- Simple ✓

Question 9:

If the cluster sizes are unequal, the mean of cluster size is equal to

$$\bar{M} = \bar{M}'N$$

- False ✓
- True

Question 10:

Mrs. Trahan samples her class by selecting 5 girls and 7 boys. This type of sampling is called?

- Stratified ✓
- Systematic
- Cluster
- Simple

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Note: Live quiz MCQs also repeated in Exam (6, 7)



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